

Dongwook Kwon

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EDUCATION

Kwangwoon University <i>Master of Science in Computer Engineering, GPA 4.33 / 4.5 (98.1%)</i>	Seoul, Korea Mar. 2024 – Feb. 2026
University of Toronto <i>Graduate Research Program, Mechanical & Industrial Engineering, GPA 3.68 / 4.0</i>	Toronto, ON, Canada Jan. 2025 – Jul. 2025
Kwangwoon University <i>B.S. Electronics & Communications Engineering, GPA 3.41 / 4.5 (88.1%)</i>	Seoul, Korea Mar. 2019 – Feb. 2024

EXPERIENCE

AI Engineer, Intelligent Agent Technology Team <i>Suresoft Technologies</i>	Dec. 2025 – Present Seongnam, Korea
- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification. - Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale. - Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.	
Research Assistant <i>Bio-Computing and Machine Learning Laboratory, Kwangwoon University</i>	Aug. 2021 – Dec. 2025 Seoul, Korea
- Researched deep-learning algorithms for anomaly detection in time-series data (ECG, cyber-physical systems). - Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices. - Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.	
Visiting Researcher <i>LG Electronics Canada (with University of Toronto)</i>	Jan. 2025 – Jul. 2025 Toronto, ON, Canada
- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems. - Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.	
AI Development Project Participant, Summer Program <i>Qualcomm Institute, University of California San Diego</i>	Jul. 2022 – Aug. 2022 San Diego, CA, USA
- Built a personality-based drug-addiction prediction system using machine-learning models. - Recognized with the program's Achievement Award.	

PROJECTS

Agentic AI Evaluation Framework <i>Python, LLMs, multi-agent systems</i>	2025 – Present
Developed an evaluation framework for agentic AI systems leveraging LLMs to assess performance and reliability.	
AI Intrusion Detection for APR1400 Nuclear Reactors <i>Python, PyTorch, anomaly detection</i>	2022 – Present
Built an AI-based defense system protecting nuclear power plant control networks from cyber attacks.	
Quadruped Robot with Vision SLAM <i>ROS, depth camera, SLAM</i>	Apr 2022 – Nov 2022
Developed autonomous navigation algorithm with SLAM and depth-camera perception for a quadruped platform.	
Smart Logistics with Computer Vision <i>PyTorch, OpenCV, container tracking</i>	Apr 2021 – Nov 2021
Logistics detection system with location tracking and stack-layer identification — Ministerial Gold Award.	

PUBLICATIONS & PATENTS

D. Kwon[†] et al. “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades.” *ACL 2026 — Industry Track (Poster)*. [†]Equal contribution.

D. Kwon[†] et al. “GEMMAS: Graph-based Evaluation Metrics for Multi-Agent Systems.” *EMNLP 2025 — Industry Track (Oral)*. [†]Equal contribution.

D. Kwon et al. “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments.” Under review at *DMKD* (Springer Nature), 2025.

D. Kwon et al. “Hybrid GAT + Transformer for implantable-device anomaly detection.” *IBEC 2024*. **Best Poster Silver**.

Patent **KR 10-2961984** — Real-time anomaly detection for implantable medical devices (Apr 2026).

Patent **KR 10-2848814** — GPT API-Based Network Anomaly Detection (Aug 2025).

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB, Bash

Frameworks: PyTorch, TensorFlow, Transformers, LangChain, NumPy, Pandas, scikit-learn

Developer Tools: Linux, Docker, Git, CUDA, VS Code

Domains: Agentic AI, multi-agent LLM systems, GNNs, time-series anomaly detection, biomedical signals